ON A PUNCTURED SPECTRUM THE COHOMOLOGY OF THE DERIVED I-ADIC COMPLETION OF RGAMMA EQUALS THE LIMIT OF THE COHOMOLOGY OF THE REDUCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3735-3746 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 3748-3766, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al01 batch). Spans per computed label truth (al_clean_queue), not pool-card lines.

Editorial note. Cross-chapter homonym warning recorded in dedup (chapter-aware gate).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name inside the retained author proof become stub prereqs.

Source reference. Original lemma 0D60: sources/stacks-algebraization/pinned-original-source.tex, source lines 3766–3766. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DX1: sources/stacks-algebraization/pinned-original-source.tex, source lines 881–893. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A0K: sources/stacks-algebraization/pinned-original-source.tex, source lines 1902–1909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BLX: sources/stacks-algebraization/pinned-original-source.tex, source lines 1777–1798. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:algebraization.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let U be the punctured spectrum of a Noetherian local ring A. Let $\mathcal{F}$ be a coherent $\mathcal{O}_U$-module. Let $I \subset A$ be an ideal. Then*

$$H^i(R\Gamma(U, \mathcal{F})^\wedge) = \lim H^i(U, \mathcal{F}/I^n\mathcal{F})$$

for all i where $R\Gamma(U, \mathcal{F})^\wedge$ denotes the derived I-adic completion.

Proof. By Lemmas 0BLX and 0A0K we have

$$R\Gamma(U, \mathcal{F})^\wedge = R\Gamma(U, \mathcal{F}^\wedge) = R\Gamma(U, R\lim \mathcal{F}/I^n\mathcal{F})$$

Thus we obtain short exact sequences

$$0 \to R^1 \lim H^{i-1}(U, \mathcal{F}/I^n\mathcal{F}) \to H^i(R\Gamma(U, \mathcal{F})^\wedge) \to \lim H^i(U, \mathcal{F}/I^n\mathcal{F}) \to 0$$

by Cohomology, Lemma 0D60. The R^1 lim terms vanish because the inverse systems of groups $H^i(U, \mathcal{F}/I^n\mathcal{F})$ satisfy the Mittag-Leffler condition by Lemma 0DX1. □